

VIWA, Fiji

WMO: 916700

Lat: 17.150S

Lon: 176.900E

Elev: 5

StdP: 101.26

Time Zone: 12.00 (PCF)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.2	22.3	14.8	10.5	23.2	15.9	11.3	23.9	13.1	24.0	11.7	23.8	2.8	80	0.373

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.6	32.5	26.9	31.8	26.7	31.3	26.4	28.2	31.1	27.7	30.6	27.4	30.2	2.1	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
27.2	23.0	30.2	26.9	22.6	29.7	26.6	22.1	29.3	90.6	31.3	88.6	30.8	87.0	30.3	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	9.0	7.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.0	28.2	28.3	28.2	27.6	26.7	26.4	25.5	25.4	26.1	26.6	27.3	28.0
	DBStd	1.57	1.19	1.09	1.05	1.29	1.87	1.15	0.99	1.08	1.11	1.10	1.08	1.09
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	1	0	0	0	0	1	0	0	0	0	0	0	0
	CDD10.0	6212	564	513	564	528	517	492	481	478	482	515	519	558
	CDD18.3	3171	306	280	306	278	259	242	223	220	232	257	269	299
	CDH23.3	30438	3504	3271	3483	2940	2437	1989	1462	1396	1749	2216	2682	3310
(14) Wind	WSAvg	3.1	2.7	2.5	2.4	2.5	2.8	3.6	3.9	4.2	3.5	3.0	2.7	2.9
	PrecAvg	1895	237	199	220	219	175	120	82	72	105	146	143	176
	PrecMax	2732	553	374	433	640	483	300	172	194	289	325	383	429
	PrecMin	1300	15	25	53	9	23	17	12	9	5	28	28	16
	PrecStd	361	131	94	94	116	111	70	45	42	59	83	96	83
	DB	33.5	33.1	32.8	32.1	31.2	30.8	30.0	30.0	30.9	31.6	32.7	32.7	32.7
	MCWB	27.7	27.3	27.3	27.2	26.1	25.7	24.6	25.0	25.1	25.1	25.9	26.7	26.7
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	32.4	32.1	31.7	31.3	30.4	29.8	29.0	28.9	29.9	30.4	31.2	31.7	31.7
	2% MCWB	27.0	27.1	27.0	26.5	25.6	25.1	24.0	24.5	24.5	24.9	25.4	26.4	26.4
	5% DB	31.5	31.5	31.1	30.7	29.6	29.1	28.2	28.2	29.1	29.5	30.4	31.1	31.1
	5% MCWB	26.7	26.8	26.7	26.1	25.0	24.7	23.5	23.9	24.1	24.6	25.4	26.2	26.2
	10% DB	30.9	30.8	30.6	30.1	29.0	28.4	27.5	27.5	28.3	28.8	29.7	30.5	30.5
	10% MCWB	26.5	26.5	26.5	25.9	24.6	24.3	23.1	23.4	23.8	24.4	25.3	25.9	25.9
	WB	28.6	28.7	28.5	28.2	27.1	27.0	26.1	26.2	26.4	26.5	27.5	28.2	28.2
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	32.2	31.4	31.3	31.5	29.7	29.4	28.4	28.6	29.2	29.4	30.5	31.0	31.0
	2% WB	27.9	28.0	27.9	27.3	26.5	26.1	25.2	25.4	25.7	25.9	26.9	27.5	27.5
	2% MCDB	30.9	31.0	30.7	30.1	29.0	28.5	27.3	27.7	28.2	29.0	29.6	30.4	30.4
	5% WB	27.5	27.6	27.4	26.9	26.1	25.6	24.6	24.7	25.1	25.4	26.5	27.0	27.0
	5% MCDB	30.4	30.4	30.2	29.5	28.5	27.9	26.9	27.0	27.6	28.3	29.1	29.9	29.9
	10% WB	27.1	27.1	27.0	26.4	25.6	25.1	24.1	24.2	24.6	24.9	26.0	26.6	26.6
	10% MCDB	29.9	29.8	29.7	29.1	28.0	27.4	26.7	26.5	27.2	27.7	28.6	29.4	29.4
(35) Mean Daily Temperature Range	MDBR	4.7	4.6	4.6	4.7	4.4	4.0	4.0	4.0	4.5	4.5	4.8	4.7	4.7
	5% DB	6.0	5.7	5.4	5.6	5.3	4.8	4.9	4.9	5.6	5.5	5.8	5.6	5.6
	MCWBR	2.7	2.6	2.6	2.7	2.6	2.3	2.3	2.3	2.6	2.5	2.6	2.6	2.6
	5% WB	5.2	5.2	5.1	5.0	4.5	4.5	4.4	4.6	5.0	4.9	4.9	4.9	4.9
	MCWBR	2.7	2.6	2.7	2.7	2.4	2.4	2.3	2.3	2.6	2.5	2.5	2.6	2.6
(40) Clear-Sky Solar Irradiance	taub	0.393	0.386	0.375	0.375	0.370	0.369	0.365	0.365	0.375	0.380	0.382	0.380	0.380
	taud	2.608	2.638	2.690	2.677	2.659	2.646	2.642	2.645	2.613	2.616	2.629	2.650	2.650
	Ebn at Noon	952	948	938	902	872	855	869	899	923	943	956	963	963
	Edh at Noon	104	100	92	87	83	82	84	89	97	101	101	99	99
(44) All-Sky Solar Radiation	RadAvg	6.33	6.03	5.57	4.90	4.33	4.05	4.26	4.78	5.50	5.95	6.28	6.41	6.41
	RadStd	0.67	0.52	0.43	0.37	0.35	0.25	0.22	0.44	0.42	0.42	0.61	0.34	0.34

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.20	+0.23	N/A	N/A	+60	+60

Nomenclature: See separate page